

KHONUU, Russian Federation

WMO: 243820

Lat: 66.450N

Lon: 143.233E

Elev: 643

StdP: 14.36

Time Zone: 11.00 (E11)

Period: 09-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-60.4	-57.3	-62.9	0.1	-59.5	-60.0	0.2	-57.2	14.4	-9.8	11.6	-7.4	0.0	N/A	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.1	83.7	63.4	78.9	60.8	74.6	59.4	65.5	80.0	63.1	76.0	60.8	71.8	5.1	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
60.0	79.2	69.7	57.7	72.7	67.6	55.5	67.2	65.8	30.7	80.7	29.0	76.4	27.3	72.2	73.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.3	13.5	11.7	-62.1	91.0	2.3	4.2	-63.8	94.0	-65.1	96.5	-66.4	98.8	-68.1	101.8		
(5)			-62.1	70.0	2.3	2.3	-63.8	71.7	-65.1	73.0	-66.4	74.3	-68.1	76.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.0	-39.9	-35.9	-13.1	16.8	40.5	57.4	60.3	54.3	39.0	10.0	-21.3	-38.6		
	DBStd	39.18	11.97	11.68	14.08	12.39	9.77	7.38	7.94	7.35	8.40	12.94	13.27	10.40		
	HDD50	15009	2788	2404	1955	997	323	20	7	42	344	1241	2140	2748		
	HDD65	19770	3253	2824	2420	1447	760	244	194	338	781	1706	2590	3213		
	CDD50	784	0	0	0	0	28	242	325	175	14	0	0	0		
	CDD65	70	0	0	0	0	0	17	47	6	0	0	0	0		
	CDH74	1086	0	0	0	0	7	273	659	147	0	0	0	0		
	CDH80	337	0	0	0	0	1	54	254	28	0	0	0	0		
	WSAvg	3.4	0.6	0.7	3.0	4.8	6.0	5.8	5.1	4.8	4.6	3.5	1.1	0.6		
	PrecAvg	7.7	0.3	0.3	0.2	0.2	0.4	1.1	1.6	1.6	0.8	0.5	0.4	0.3		
(15) Precipitation	PrecMax	10.7	0.9	0.8	0.7	0.6	1.3	2.7	3.4	4.2	2.2	1.2	1.3	0.6		
	PrecMin	5.2	0.1	0.0	0.0	0.0	0.0	0.2	0.2	0.3	0.0	0.0	0.1	0.0		
	PrecStd	1.5	0.2	0.2	0.2	0.1	0.4	0.6	0.7	0.9	0.5	0.3	0.2	0.2		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-3.8	0.5	29.7	47.7	74.3	86.2	93.8	84.3	70.4	39.7	28.9	-5.5		
		MCWB	-4.6	-0.7	24.2	37.1	54.1	62.3	67.2	65.2	54.5	32.1	24.3	-6.2		
	2%	DB	-8.6	-7.2	24.0	43.4	68.1	80.7	87.0	78.0	65.0	33.4	8.8	-11.8		
		MCWB	-9.1	-7.7	19.4	34.5	51.2	59.6	65.3	60.9	50.9	29.0	7.5	-12.2		
	5%	DB	-15.7	-12.7	18.4	40.4	63.0	77.0	81.4	73.8	60.0	30.3	1.7	-18.4		
		MCWB	-16.0	-13.1	15.2	32.4	48.6	58.1	63.7	59.1	48.5	27.7	1.2	-18.8		
	10%	DB	-21.6	-18.0	12.1	36.9	58.2	73.0	76.1	69.6	54.1	27.5	-2.3	-22.5		
		MCWB	-21.8	-18.4	9.7	29.9	45.7	57.1	61.9	57.8	44.8	25.4	-2.9	-22.7		
	0.4%	WB	-4.4	-0.5	24.7	37.7	54.6	64.4	69.5	66.6	55.3	33.3	24.4	-6.1		
		MCDB	-4.1	0.5	30.1	47.1	71.5	80.7	87.1	81.5	69.0	38.1	29.3	-5.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	-8.8	-7.7	19.7	34.8	51.9	61.7	66.6	62.8	51.9	30.2	7.8	-12.0		
		MCDB	-8.2	-7.1	23.7	43.0	65.7	77.2	82.3	75.3	63.7	33.0	8.8	-11.6		
	5%	WB	-16.0	-13.1	15.0	32.7	49.3	59.5	64.7	60.4	49.0	27.8	1.1	-18.6		
		MCDB	-15.8	-12.8	18.4	39.8	62.0	73.8	78.0	70.6	58.6	30.2	1.5	-18.4		
	10%	WB	-21.8	-18.3	9.8	30.1	46.4	57.7	62.7	58.3	46.0	25.4	-2.9	-22.6		
		MCDB	-21.7	-17.9	12.2	36.4	57.1	71.1	75.0	67.6	53.7	27.4	-2.3	-22.4		
	5% DB	MDBR	11.3	17.8	29.7	26.2	19.8	20.5	19.1	20.3	17.3	14.8	12.5	9.5		
		MCDBR	16.8	19.4	30.8	24.1	26.9	28.1	30.7	30.5	29.0	15.1	16.5	16.1		
		MCWBR	16.5	18.8	27.0	17.3	13.8	12.2	13.2	15.7	16.8	11.7	15.5	16.0		
		MCWBR	16.9	19.0	30.3	22.9	25.8	26.0	25.9	25.7	26.7	11.7	16.3	16.2		
(35) Mean Daily Temperature Range	5% WB	MCWBR	16.6	18.5	26.7	16.6	13.1	12.6	12.3	13.8	15.8	9.1	15.4	16.1		
	Clear-Sky Solar Irradiance	taub	0.161	0.227	0.286	0.337	0.378	0.374	0.448	0.394	0.309	0.250	0.171	0.165		
		taud	1.917	2.103	2.086	2.038	2.193	2.390	2.162	2.309	2.472	2.240	1.935	1.914		
(42) All-Sky Solar Radiation	RadAvg	Ebn at Noon	88	203	241	256	258	264	235	238	241	192	74	9		
		Edh at Noon	5	16	29	40	38	32	40	31	20	14	4	1		
(44) All-Sky Solar Radiation	RadStd		27	248	814	1515	1761	1730	1544	1171	683	320	62	3		
			2	19	34	48	67	124	168	91	61	34	4	0		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page